YILAN ZHENG

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EDUCATION

Duke University — **Durham, NC**
PhD student, Finance — *August 2021 – Current*

University of Chicago — **Chicago, IL**
Master of Arts, Economics — *September 2018 – June 2019*

University of International Business and Economics — **Beijing, China**
Bachelor of Arts, Economics and Finance — *September 2014 – July 2018*

PROFESSIONAL EXPERIENCE

OpenBlock Labs — **Palo Alto, CA**
Data Scientist — *February 2024 – Current*

- Specialized in mechanism design and implementation of incentive models for blockchain protocols. Developed key market metrics and utilized these metrics along with exploratory quantitative research and blockchain data to create and adjust models, optimizing market efficiency and incentive distribution.
- Developed advanced predictive models using time series and machine learning techniques and conducted comprehensive statistical analysis to forecast key market performance indicators, including transaction volume and points accrued.
- Developed ETL pipelines and RESTful APIs using FastAPI, leveraging AWS technologies such as Athena, Glue, S3, and data warehouse solutions like Snowflake to optimize data handling and storage capabilities.
- Managed product development and data processing workflows for data-driven projects, utilizing Airflow to deliver key outcomes in incentive allocation and market performance predictions.

Huobi Global — **Hong Kong, HK**
Quantitative Researcher Intern — *July 2021 – September 2021*

- Analyzed the regime switching in the cryptocurrency market, employ regression analysis, time series analysis, and random forest to predict market performance using data from multiple exchanges and mining operations.
- Assisted in the analysis and the design of crypto derivative products such as callable bull/bear contracts.
- Compiled and summarized daily market-making data, performing detailed analysis of market volume, profits, and trends to inform strategic decisions.

Columbia Business School — **New York, NY**
Research Data Scientist — *July 2019 – June 2021*

- Conducted advanced data manipulation and empirical analysis with Python (pandas) and R (data.table and dplyr), effectively delivering insights via reports and visualization tools such as ggplot and matplotlib.
- Managed and synthesized large, high-frequency datasets from observational and survey sources using SQL, ensuring efficient database maintenance across multiple servers and organizations.
- Engaged in diverse projects in fintech, financial intermediation and consumer finance, including:
  - Investigating the growth of fintech and non-fintech shadow banks in loan originations using propensity score matching to understand regulatory and technological impacts.
  - Studying the industrial organization of financial intermediation and the influence of bank business models on mortgage origination using regression discontinuity design and structural estimation.
  - Predicting mortgage loan delinquency and default using machine learning techniques.

Fuqua School of Business, Duke University — **Durham, NC**
Graduate Student Researcher — *May 2022 – Dec 2023*

- Independent Research Project on Vote-Escrow mechanism: Developed and analyzed a two-sided market model for DEX liquidity provision efficiency with IV regression and random coefficient models using self-scraped data from Curve.
- Independent Research Project on Mortgage Response to TCJA: Investigated household mortgage response to monetary incentives using natural experiment through regression kink design and McCrary test for robustness.
- Scraped, compiled, and maintained a geodatabase with emissions, firm locations, and monitoring stations; analyzed the impact of environmental policies on firm performance using causal analysis.
- Conducted a comprehensive study on cryptocurrency perpetual swaps, utilizing Tardis API and data self-scraped from multiple exchanges. Focused on constructing index prices and analyzing the determination of funding rates.

SKILLS

- Programming Languages: Python, R, SQL, Stata, Pytorch, Tensorflow, Matlab, Latex, JavaScript, GIS
- Technical Skills: causal inference, time series, machine learning, experimental design, NLP